

Final

- 2026/05/08 (FRI) 19:00–22:00 澳門理工大學體育館
- 5 short-answer questions and 2 calculation questions
- 带计算器!

1. Basic knowledge of R operator

a. Arithmetic Operators

算数运算符： Used to perform basic mathematical calculations

运算符	解释	示例	输出
x <- 5, y <- 16			
+	Addition 加	x + y	21
-	Subtraction 减	x - y	-11
*	Multiplication 乘	x * y	80
/	Division 除	y / x	3.2
^	Exponentiation 幂运算	y ^ x	1048576
%%	Modulus 取余	y %% x	1
%/%	Integer division 取整	5 %/% 2	2

b. Relational Operators

关系运算符： Used to compare values and return logical results (TRUE or FALSE).

运算符	解释	示例	输出
x <- 5, y <- 16			
<	Less than	x < y	TRUE
>	Greater than	x > y	FALSE
<=	Less than or equal to	x <= 5	TRUE

<code>>=</code>	Greater than or equal to	<code>y >= 20</code>	FALSE
<code>==</code>	Equal to	<code>y == 16</code>	TRUE
<code>!=</code>	Not equal to	<code>x != 5</code>	FALSE

c. Logical Operators

逻辑运算符: Used to perform logical operations such as NOT, AND, and OR, returning Boolean values (TRUE or FALSE).

- **AND**: 两个条件都必须为真，结果才会是 TRUE; 如果其中一个是 FALSE，结果就一定是 FALSE。
- **OR**: 只要有一个条件为 TRUE，返回 TRUE

运算符	解释	示例	输出
	0 被当作 FALSE 非 0 被当作 TRUE	<code>x <- c (TRUE, FALSE, 0, 6)</code> <code>y <-c (FALSE, TRUE, FALSE, TRUE)</code>	
<code>!</code>	Logical NOT, Get the inverse value	<code>!x</code>	FALSE TRUE TRUE FALSE
<code>&</code>	Element-wise logical AND 逐一元素	<code>x & y</code>	FALSE FALSE FALSE TRUE
<code>&&</code>	Logical AND 只检查第一个元素	<code>x && y</code>	FALSE
<code> </code>	Element-wise logical OR 逐一元素	<code>x y</code>	TRUE TRUE FALSE TRUE
<code> </code>	Logical OR 只检查第一个元素	<code>x y</code>	TRUE

【拓展】 `&&` 和 `||` 只检查第一个元素

- 在 R 里，很多对象都是 向量，所以像 `&` 和 `|` 这样的运算符会逐元素比较，返回一个向量结果。
- 但是在 控制语句（比如 `if`、`while`）里，条件只能是一个单一的布尔值（TRUE 或 FALSE），不能是一个向量。
 - `&&` 和 `||` 的设计目的就是：只检查第一个元素，返回一个单一布尔值。
 - 直接用于控制语句，不会因为返回一个向量而出错。

d. Assignment Operators

赋值运算符: Used to assign values to variables in R, either leftwards or rightwards, and sometimes across environments.

分类	运算符	解释
Leftwards assignment	<code><-</code> / <code>=</code>	二者基本等价，常用。箭头指向的是变量名（被赋值的对象） Same environment
	<code><<-</code>	Parent/global environment
Rightwards assignment	<code>->></code>	赋值到父环境或全局环境，可以修改函数外部的变量
	<code>-></code>	语法上可用，但实际很少使用，箭头指向的是变量名（被赋值的对象） Same environment

Additional content

(1) Operation on vector

In R, all operations are carried out in element-wise fashion. If vectors have different lengths, R uses recycling to match lengths.

1. 基本向量运算

```
>x <- c(2,8,3)
>y <- c(6,4,1)
>x+y
>[1] 8    12   4
>x>y
>[1] FALSE TRUE TRUE
```

2. 向量长度不一致时的循环补齐 (Recycling Rule)

```
>x <- c(2,1,8,3)
>y <- c(9,4)
>x+y # Element of y is recycled to 9,4,9,4
>[1] 11 5 17 7
```

3. 标量与向量运算

▼

R |

```
1 x <- c(2,1,8,3)
2 x - 1 # Scalar 1 is recycled to 1,1,1,1
3 # [1] 1 0 7 2
```

- 在 R 里，`x - 1` 的逻辑是：
 - `1` 被当作一个长度为 1 的向量。
 - R 会自动 循环补齐，变成 `1,1,1,1`，然后和 `x` 的每个元素逐一相减。

- 所以结果是：(2-1, 1-1, 8-1, 3-1)=1, 0, 7, 2

4. 非整倍数长度的警告

- 警告 (Warning) → 程序继续跑，有结果，但提醒注意可能的逻辑问题。
- 安全性考虑：如果长向量的长度不是短向量的整数倍，补齐后最后一轮会“不完整”。
 - 比如 `x` 长度是 4，`c(1,2,3)` 长度是 3。
 - 补齐后是 `1,2,3,1`，但其实只补了 3 个完整元素 + 1 个额外元素。
 - R 担心你可能是 无意写错，所以会给出警告，提醒你注意长度不匹配。

```

1 x <- c(2,1,8,3)
2 # 非整倍数补齐：长度为 3，不能整倍数补齐到 4
3 x + c(1,2,3)
4 # [1] 3 3 11 4
5 # Warning: longer object length is not a multiple of shorter object length

```

(2) Exercise

```

1 x <- c(1:10)
2 x[(x > 8) | (x < 5)]
3 # 结果是 1 2 3 4 9 10

```

- `x <- c(1:10)`：生成一个向量 `1,2,3,...,10`。
- `(x > 8)`：逻辑判断，得到 `FALSE FALSE ... TRUE TRUE`（只有 9 和 10 大于 8）。
- `(x < 5)`：逻辑判断，得到 `TRUE TRUE TRUE TRUE FALSE ...`（只有 1-4 小于 5）。
- `(x > 8) | (x < 5)`：逻辑或，结果是前 4 个和最后 2 个为 TRUE。
- `x[...]`：用逻辑向量做索引，筛选出对应位置的值 → `1 2 3 4 9 10`。

Final Question Prediction

- Short Q
- 参考 Mid-term
 - Briefly describe the four fundamental data structures in R (12 marks) .
 - Please describe at least four data types in the R. For each one, provide a simple example (12 marks)
- (List + Description + Example) * 4 = 12 marks

- List four types of operator in R and briefly explain them.

2. Hypothesis testing

Hypothesis Tests

Steps of Hypothesis Testing:

Step 1. Develop the null and alternative hypotheses.

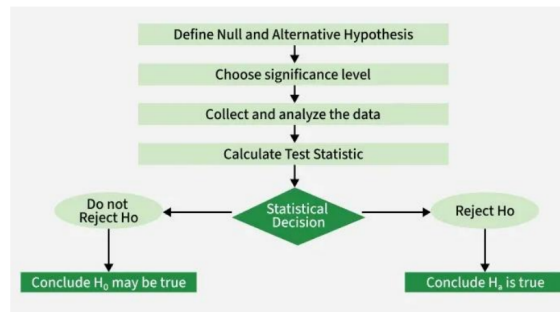
Step 2. Specify the level of significance (α).

Step 3. Collect the sample data and compute the value of the test statistic.

Step 4. Use the value of the test statistic to compute the p value.

Step 5. Reject H_0 if the $p \leq \alpha$.

Step 6. Interpret the statistical conclusion in the context of the application.



Final Question Prediction

- Short Q
- 参考 Mid-term (只有 A 卷考到了 Hypothesis testing 10 marks)

A fitness magazine advertises that the mean monthly cost of joining a health club is \$25. You want to test this claim. You find that a random sample of 18 clubs has a mean monthly cost of \$26.25 and a standard deviation of \$3.23. At $\alpha = 0.10$, do you have enough evidence to reject the advertisement's claim? **(10 marks)**

(a) The claim is "the mean monthly cost of joining a health club is \$25."

$H_0: \mu = 25$ (claim); $H_a: \mu \neq 25$ **[2 marks]**

(b) $-t_0 = -1.740$, $t_0 = 1.740$; Rejection regions: $t < -1.740$, $t > 1.740$ **[2 marks]**

(c) $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{26.25 - 25}{\frac{3.23}{\sqrt{18}}} \approx 1.642$ **[2 marks]**

(d) Because $-1.740 < t < 1.740$, fail to reject H_0 . **[2 marks]**

(e) There is not enough evidence at the 10% level of significance to reject the claim that the mean monthly cost of joining a health club is \$25. **[2 marks]**

3. Linear optimization

a. Introduction

Optimization Problems:

- Maximize or minimize **objective function**, and have a set of **constraints**.
- Can be **linear or nonlinear**.
- Typical application:
 - 制造商：制定生产计划和库存策略，既要满足未来需求，又要 **最小化生产和库存成本**。
 - 金融分析师：在股票和债券投资组合中选择方案，以 **最大化投资回报**。

Linear Optimization 线性优化 = Linear Programming 线性规划

- 通过 线性目标函数 + 线性约束条件 来求解最优解的问题

b. General Guidelines for Problem Formulation

- **Problem formulation or modeling:** Process of translating the verbal statement of a problem into a **mathematical statement/ model**
- **General guidelines for problem formulation:**
 - Understand the problem thoroughly.
 - Describe the objective.
 - Describe each constraint.
 - Define the decision variables.
 - Write the objective in terms of the decision variables.
 - Write the constraints in terms of the decision variables.

1. **Understand the problem thoroughly** 彻底理解问题，明确业务背景和限制条件。
2. **Describe the objective** 描述目标，要么最大化（如利润），要么最小化（如成本）。
3. **Describe each constraint** 描述每个约束条件，比如工时、预算或资源限制。
4. **Define the decision variables** 定义决策变量，即可控制的输入
5. **Write the objective in terms of the decision variables** 用决策变量写出目标函数
6. **Write the constraints in terms of the decision variables** 把资源限制转化为不等式。

c. Case Study

i. Background

Par, Inc. is a small manufacturer of golf equipment and supplies. Management decided to enter the market for medium- and high-priced golf bags 进入中高端高尔夫球包市场. The distributor will buy all produced bags by the end of the third month. 分销商承诺在三个月内收购所有生产的球包。

Production Operations:

- Cutting and dyeing the material 材料裁剪与染色
- Sewing 缝纫
- Finishing (adding umbrella holder, club separators, etc.) 后期加工
- Inspection and packaging 检验与包装

Production Requirements Per Bag

Department	Standard Bag	Deluxe Bag
Cutting and Dyeing	$\frac{7}{10}$	1
Sewing	$\frac{1}{2}$	$\frac{5}{6}$
Finishing	1	$\frac{2}{3}$
Inspection and Packaging	$\frac{1}{10}$	$\frac{1}{4}$

Departmental Time Availability 部门工时限制

Department	Number of hours
Cutting and Dyeing	630
Sewing	600
Finishing	708
Inspection and Packaging	135

Profit Contribution

- Standard bag: \$10 per unit
- Deluxe bag: \$9 per unit

ii. Problem Formulation

Illustration: Par, Inc. (cont.): Develop a **mathematical model** of the Par, Inc. problem to determine the **number** of standard bags and the number of deluxe bags to produce to **maximize total profit contribution**.

1. Understand the problem thoroughly

We selected the **Par, Inc. problem** to introduce linear programming.

- **Question:** How many standard bags and how many deluxe bags should be produced to maximize total profit contribution?

2. Describe the objective

- The objective is to **maximize the total contribution to profit**

3. Describe each constraint

每个部门的工时有限。约束条件保证各部门的使用时间不能超过可用时间。

Constraint	Description
1	Number of hours of cutting and dyeing time used must be less than or equal to the number of hours of cutting and dyeing time available.
2	Number of hours of sewing time used must be less than or equal to the number of hours of sewing time available.
3	Number of hours of finishing time used must be less than or equal to the number of hours of finishing time available.
4	Number of hours of inspection and packaging time used must be less than or equal to the number of hours of inspection and packaging time available.

1. Cutting & Dyeing (裁剪与染色)

$$0.7S + 1D \leq 630$$

2. Sewing (缝纫)

$$0.5S + 0.83D \leq 600$$

3. Finishing (后期加工)

$$1S + 0.67D \leq 708$$

4. Inspection & Packaging (检验与包装)

$$0.1S + 0.25D \leq 135$$

4. Define the decision variable

- Define the **decision variables**: The controllable inputs for Par, Inc. are:
 - S = number of standard bags.
 - D = number of deluxe bags.
- In optimization terminology, S and D are referred to as the decision variables.

5. Write the objective in terms of the decision variables

- Write the objective in terms of the decision variables.
 - Par's profit contribution comes from two sources: the profit contribution made by S and D produced.
 - If Par makes \$10 for every standard and \$9 for every deluxe bag,
Total profit contribution = $10S + 9D$ = objective function.
- Objective: Max $10S + 9D$.

6. Write the constraints in terms of the decision variables

- Constraint 1: $\left(\begin{array}{c} \text{Hours of cutting and} \\ \text{dyeing time used} \end{array} \right) \leq \left(\begin{array}{c} \text{Hours of cutting and} \\ \text{dyeing time available} \end{array} \right)$

Production Time (hours)		
Department	Standard Bag	Deluxe Bag
Cutting and Dyeing	$\frac{7}{10}$	1
Sewing	$\frac{1}{2}$	$\frac{5}{6}$
Finishing	1	$\frac{2}{3}$
Inspection and Packaging	$\frac{1}{10}$	$\frac{1}{4}$

Total hours of cutting and dyeing time used
 $= 7/10S + 1D$

Department	Number of hours
Cutting and Dyeing	630

$$\frac{7}{10}S + 1D \leq 630$$

- Constraint 2: $\left(\begin{array}{c} \text{Hours of sewing} \\ \text{time used} \end{array} \right) \leq \left(\begin{array}{c} \text{Hours of sewing} \\ \text{time available} \end{array} \right)$

Production Time (hours)		
Department	Standard Bag	Deluxe Bag
Cutting and Dyeing	$\frac{7}{10}$	1
Sewing	$\frac{1}{2}$	$\frac{5}{6}$
Finishing	1	$\frac{2}{3}$
Inspection and Packaging	$\frac{1}{10}$	$\frac{1}{4}$

Total hours of sewing time used $= 1/2S + 5/6D$

Department	Number of hours
Sewing	600

$$\frac{1}{2}S + \frac{5}{6}D \leq 600$$

- Constraint 3: $\left(\begin{array}{c} \text{Hours of finishing} \\ \text{time used} \end{array} \right) \leq \left(\begin{array}{c} \text{Hours of finishing} \\ \text{time available} \end{array} \right)$

Production Time (hours)		
Department	Standard Bag	Deluxe Bag
Cutting and Dyeing	$\frac{7}{10}$	1
Sewing	$\frac{1}{2}$	$\frac{5}{6}$
Finishing	1	$\frac{2}{3}$
Inspection and Packaging	$\frac{1}{10}$	$\frac{1}{4}$

Total hours of finishing time used $= 1S + 2/3D$

Department	Number of hours
Finishing	708

$$1S + \frac{2}{3}D \leq 708$$

- Constraint 4: $\left(\begin{array}{c} \text{Hours of inspection and} \\ \text{packaging time used} \end{array} \right) \leq \left(\begin{array}{c} \text{Hours of inspection and} \\ \text{packaging time available} \end{array} \right)$

Production Time (hours)		
Department	Standard Bag	Deluxe Bag
Cutting and Dyeing	$\frac{7}{10}$	1
Sewing	$\frac{1}{2}$	$\frac{5}{6}$
Finishing	1	$\frac{2}{3}$
Inspection and Packaging	$\frac{1}{10}$	$\frac{1}{4}$

Total hours of inspection and packaging time used
 $= 1/10S + 1/4D$

Department	Number of hours
Inspection and Packaging	135

$$\frac{1}{10}S + \frac{1}{4}D \leq 135$$

- **Nonnegativity constraints:** Based on the fact that the number of standard or deluxe bags produced **cannot be negative**. $S \geq 0$ and $D \geq 0$ or $S, D \geq 0$

iii. Mathematical model

Mathematical Model for the Par, Inc. Problem:

- We succeeded in translating the objective and constraints of the problem into a set of mathematical relationships, referred to as a mathematical model.
- **Mathematical model:** A set of mathematical relationships

Max $10S + 9D$.	subject to (s.t.)	$\frac{7}{10}S + 1D \leq 630$ Cutting and dyeing
		$\frac{1}{2}S + \frac{5}{6}D \leq 600$ Sewing
		$1S + \frac{2}{3}D \leq 708$ Finishing
		$\frac{1}{10}S + \frac{1}{4}D \leq 135$ Inspection and packaging
		$S, D \geq 0$

Our job now is to find the product mix (the combination of values for S and D) that satisfies all the constraints and at the same time yields the highest objective function value. 既满足所有约束条件，又能使目标函数的值达到最大。

This is a linear programming model because both the objective function and all constraint functions are linear functions of the decision variables. 这是一个线性规划模型，因为目标函数和所有约束条件都是决策变量的线性函数。

A linear function means each variable appears in a separate term and is raised to the first power. 线性函数的特点是：每个变量单独出现，并且指数为 1。

Linear programming model

线性规划模型 Linear programming model 成立需要满足三个假设 assumptions:

1. **Proportionality 比例性** : The contribution to the objective function and the resources used in each constraint are proportional to the value of each decision variable. 每个决策变量对目标函数的贡献，以及在约束中消耗的资源，都与该变量的取值成比例。
 - 做多少产品，就贡献多少利润、消耗多少资源，线性关系。
 - 举例：做 1 个标准包赚 \$10，做 2 个标准包就赚 \$20，做 3 个就是 \$30 ——利润和数量成比例。
2. **Additivity 可分性** : The total value of the objective function and resources used can be found by summing the contributions of all decision variables. 目标函数的总值和资源消耗可以通过所有决策变量的贡献相加得到
 - 不同产品的贡献可以直接相加。
 - 举例：做 2 个标准包赚 \$20，做 3 个豪华包赚 \$27，总利润就是 \$47，直接相加，不会有额外复杂的交互作用。
3. **Divisibility 可加性** : The decision variables are continuous, meaning they can take any value greater than or equal to zero (together with non-negativity constraints). 决策变量是连续的，可以取任何大于或等于零的值（结合非负性约束）
 - 决策变量可以是连续的，不一定只能是整数。

- 举例：模型里可以出现“生产 2.5 个标准包”，虽然现实中不可能半个包，但数学模型允许这样计算，方便用线性规划方法求解。

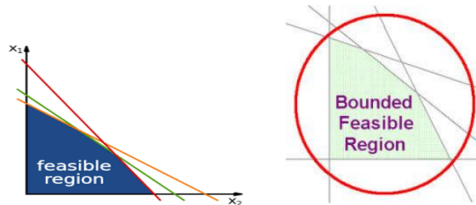
总而言之，

- 比例性：一件一件地增加，贡献和消耗成比例。
- 可加性：不同产品的贡献可以直接加在一起。
- 可分性：变量可以是连续值，不必限制为整数。

iv. Solving the Par, Inc. Problem

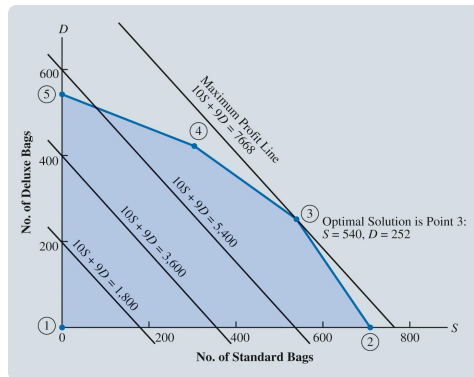
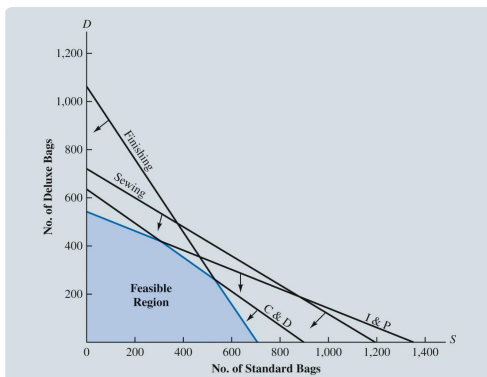
- The optimal solution must have the **highest objective function value**.
- The optimal solution must be feasible (**satisfy all constraints**).
- Search over the **feasible region** (set of all possible solutions). 在可行域中寻找最优解：在所有满足条件的解里，找到利润最大的那个点

Graphing the **feasible region** and discussing the geometry of the feasible region of the model will help us **better understand** linear programming



Geometry of the Problem

- 只要每个决策变量都是一次方，并且没有相乘或平方项，这个函数就是线性的
- If constraints are **inequalities**, each constraint cuts the space into two **half spaces**. 每个不等式约束会把空间切成两个半空间
 - The intersection of these half spaces forms the **feasible region**. 所有半空间的交集就是可行域。
 - 可行域通常是一个封闭的多边形，最优解一定在它的顶点。



- Feasible Region

- 阴影区域就是所有约束条件的交集，区域内的每个点都满足所有工时限制 **constraint** 和非负性 **nonnegativity**。
- 可行域就是“所有条件都满足的解的集合”，在图上表现为一个多边形区域。

• Optimal Solution

- The optimal solution is the point in the feasible region that yields the highest possible objective function value.
- **Contour lines** 等值线 represent sets of points with the same objective function value. 目标函数取相同值的所有点
- The highest contour line **intersecting** the feasible region gives the optimal solution.
- For Par, Inc., the highest contour line is: $10S + 9D = 7668$, and it touches the feasible region at **point 3**.
- Based on the geometry of Figure, To solve a **linear** optimization problem we only have to search the **extreme points** of the **feasible region** to find the **optimal solution**.
 - Extreme points are found where **constraints intersect** on the boundary of the **feasible region**.
 - To find the extreme point:
 - Each extreme point **lies at the intersection of two constraint lines**
 - 把约束条件当作直线，交点就是候选解

The values of S and D and the objective function value are:

- The **highest profit** is achieved at **point 3**.
- The **optimal plan** is to produce 540 Standard bags and 252 Deluxe bags.

Point	S	D	Profit = $10S + 9D$
1	0	0	$10(0) + 9(0) = 0$
2	708	0	$10(708) + 9(0) = 7,080$
3	540	252	$10(540) + 9(252) = 7668$
4	300	420	$10(300) + 9(420) = 6,780$
5	0	540	$10(0) + 9(540) = 4,860$

v. Alternative Optimal Solutions

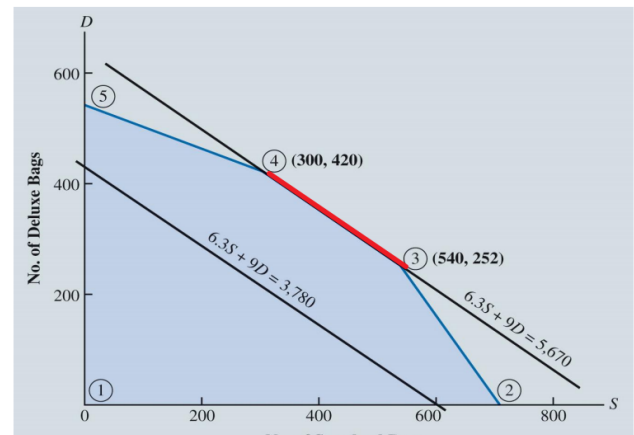
多重最优解: An **alternative optimal solution** is one in which the **optimal objective function contour line coincides with** one of the binding **constraint lines** on the boundary of the feasible region. 当目标函数的等值线刚好与某个约束边界重合时

在这种情况下，不止一个点能达到相同的最优目标值。

【案例】

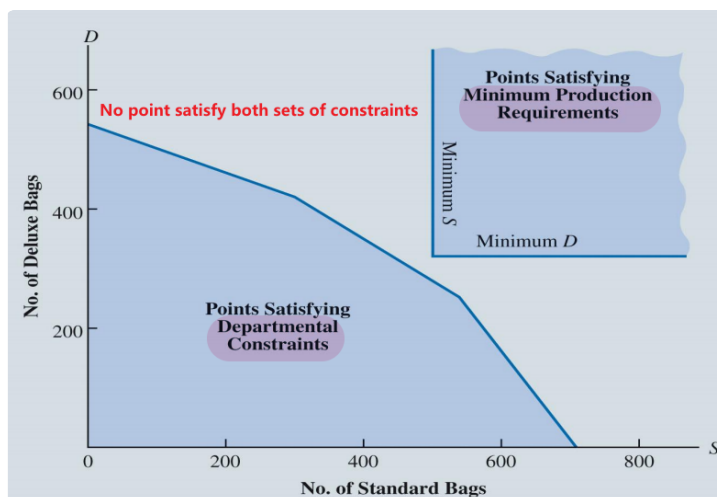
- Original objective function: $10S + 9D$.
- Assume the profit for the standard golf bag decreased to \$6.30.
- Revised objective function: $6.3S + 9D$.
- The optimal solution occurs at two extreme points:
 - Extreme point 4 ($S = 300, D = 420$).
 - Extreme point 3 ($S = 540, D = 252$).
- The objective function values at these two extreme points are identical:
 - Extreme point 4: $6.3S + 9D = 6.3(300) + 9(420) = 5670$.
 - Extreme point 3: $6.3S + 9D = 6.3(540) + 9(252) = 5670$.

The figure shows the optimal solution at two extreme points. Furthermore, any point on the line connecting the two optimal extreme points also provides an optimal solution.



vi. Infeasibility

- Infeasibility 不可行性 means no solution 无解 to the linear programming problem.
- No points satisfy all the constraints and the nonnegativity conditions simultaneously.
- Graphically, a feasible region 可行域不存在 does not exist.
- 原因:
 - Management's expectations are too high. 管理层的期望过高
 - Too many restrictions have been placed on the problem. 给问题加了太多限制条件, 导致没有解能满足所有条件



- Management should be informed that resources are not sufficient to make 500 standard bags and 360 deluxe bags.
- Provide details:
 - Minimum amounts of resources required 达到目标所需的最低资源量.
 - Current amounts available 当前实际可用的资源量.
 - Additional amounts needed to reach the target.

Operation	Minimum Required Resources (hours)	Available Resources (hours)	Additional Resources Needed (hours)
Cutting and Dyeing	$\frac{7}{10}(500) + 1(360) = 710$	630	80
Sewing	$\frac{1}{2}(500) + \frac{5}{8}(360) = 550$	600	None
Finishing	$1(500) + \frac{2}{3}(360) = 740$	708	32
Inspection and Packaging	$\frac{1}{10}(500) + \frac{1}{4}(360) = 140$	135	5

vii. Unbounded Solution

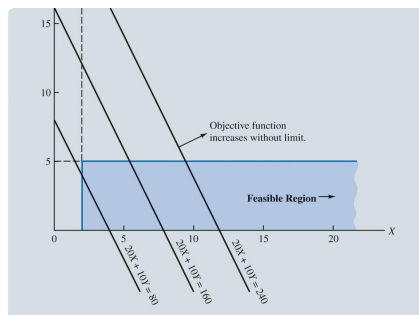
Unbounded 无界解 : The situation in which the value of the solution:

- May be made infinitely large—for a maximization problem.
- May be made infinitely small—for a minimization problem.
- Without violating any of the constraints.

解的值可以无限大（最大化问题）或无限小（最小化问题），而且不会违反任何约束条件。

- 出现的原因：The occurrence of an unbounded solution is often the result of a **missing constraint**.
 - 比如老板只说“利润越大越好”，但没有限制工时或资源，那模型就会告诉你“无限生产”，利润无限大。

$$\begin{aligned}
 &\text{Max } 20X + 10Y \\
 &\text{s.t.} \\
 &\quad 1X \geq 2 \\
 &\quad 1Y \leq 5 \\
 &\quad X, Y \geq 0
 \end{aligned}$$



- 目标函数可以无限增大 **extends indefinitely**, 因为没有限制生产数量。
- 结果：无界解。
- Reflection
 - In real-world linear programming models, an unbounded solution means the problem has been **improperly formulated 建模 不正确**.

- If a profit maximization problem results in an unbounded solution, the mathematical model does **not represent** the real-world **problem sufficiently**.
- In many cases, this error is the result of inadvertently **omitting a constraint** during problem formulation.

Final Question Prediction

- Short Q? -> 考如何建模

4. Linear regression

a. Mid-term

Q1: Describe the different variable selection procedures used in regression analysis, including their starting points and mechanisms (10 marks)

Variable selection procedures are used to determine which independent variables should be included in a regression model. They generally fall into two categories: **Iterative Procedures** and the **Best Subsets Procedure**. (2 marks)

1. Backward Elimination: (2 marks)

- **Mechanism:** It begins with a model containing **all** potential independent variables and removes them one by one based on a specific criterion. It stops when all remaining variables are significant at a specified level.
- **Use Case:** It is best used when the sample size (n) is significantly larger than the number of predictors (p).

2. Forward Selection: (2 marks)

- **Mechanism:** It begins with **no** variables in the model and adds them one by one based on a specific criterion. It stops when no remaining variables meet the threshold for addition.
- **Use Case:** It is preferred for high-dimensional data (where $p > n$) or massive datasets where computational efficiency is required.

3. Stepwise Selection: (2 marks)

- **Mechanism:** Like forward selection, it begins with **no** variables. However, it is more dynamic: in each step, it adds the most significant variable that meets the entry threshold, but then evaluates all variables currently in the model to remove the least significant one. It stops when no more variables can be added or removed.

4. Best Subsets Procedure: (2 marks)

- **Mechanism:** This is an exhaustive approach that evaluates regression models for **every possible combination** of independent variables. It starts with all simple linear regressions, moves to all two-variable combinations, and continues until all subsets are tested and compared.
-

Q2: What strategies can be employed to avoid overfitting in regression modeling, and what are the primary methods of cross-validation used to assess model performance? (10 marks)

1. Strategies to Avoid Overfitting: (4 marks)

- **Variable Selection:** Only include independent variables that are expected to have a real and meaningful relationship with the dependent variable. **(1 mark)**
- **Model Complexity:** Use complex models (such as quadratic or piecewise linear regression) only when there is a reasonable expectation that the complexity accurately depicts the underlying process. **(1 mark)**
- **Human Oversight:** Do not let software dictate the final model. Use iterative procedures like stepwise or best-subsets regression as guidance rather than as the sole method for model generation. **(1 mark)**
- **Cross-Validation:** Assess the model's performance using data other than the sample data used to build it. **(1 mark)**

2. Cross-Validation Methods:

- **Holdout Method:** The data is divided into two sets: **(2 marks)**
 - **Training Set:** Used to build candidate models.
 - **Validation Set:** Used to compare model performances and select the final model for prediction.
 - **k-fold Cross-Validation:** The sample data is randomly divided into k equal-sized, mutually exclusive, and collectively exhaustive subsets (folds). The process is repeated over k iterations. **(2 marks)**
 - **Leave-one-out Cross-Validation:** For a sample of n observations, the model is estimated using $n - 1$ observations and evaluated on the single observation that was omitted. This process is repeated for every observation in the dataset. **(2 marks)**
-

Q3: Define multicollinearity in multiple regression analysis, explain the primary problem it causes during hypothesis testing, and identify the common methods used to detect it. (10 marks)

Definition: Multicollinearity refers to the presence of correlation among the independent variables in a multiple regression model. **(2 marks)**

The Problem: It complicates the t-tests for individual parameters. Specifically, multicollinearity can lead to the conclusion that a parameter is not significantly different from zero (not statistically significant) even when the independent variable actually has a strong relationship with the dependent variable. **(4 marks)**

Detection Methods:

Correlation Coefficient: A common rule of thumb is that multicollinearity may be a problem if the absolute value of the sample correlation coefficient between any two independent variables exceeds 0.7. **(2 marks)**

VIF and Tolerance: It can also be identified using the Variance Inflation Factor (VIF) and tolerance levels. *Tolerance* should be $> .1$, and VIF should be below 10.

(2 marks)

b. Mid– term not involved

Addressing nonsignificant independent Variables 如何处理“不显著的自变量”

- If **practical experience** dictates that the nonsignificant independent variable has a relationship with the dependent variable, the independent variable should **be left** in the model.
- If the model **sufficiently explains** the dependent variable without the nonsignificant independent variable, then consider **rerunning** the regression **without** the nonsignificant independent variable.
- The appropriate treatment of the inclusion or exclusion of the y-intercept when b_0 is not statistically significant may require **special consideration**.

情况	处理方式	原因
实践经验认为变量重要	保留	现实逻辑 > 统计显著性
删除后模型仍然好	删除	简化模型、减少噪声
截距不显著	特别处理，不可随便删	删除截距会改变模型结构

Final Question Prediction

- Short Q; Linear regression 不考计算

5. Descriptive analysis: standardization and outliers

a. Measures of Location 位置度量

- **Measures of Location**: 一组统计量，用来表示数据分布的“中心”或“典型值”。
- 它们回答的问题是：**数据大致集中在哪里？**

i. Median

Median: Value in the middle when the data are arranged in ascending order.

- Odd number of observations: Middle value
- Even number of observations: Average of two middle values

ii. Mode 众数

- Value that occurs most frequently in a data set.
- 存在多个 Mode 的情况

iii. Mean

Mean/ Arithmetic Mean

SAMPLE MEAN

$$\bar{x} = \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n} \quad (2.2)$$

- 适合 **独立的、可加的数据**，比如分数、温度、收入金额。
- 每个数据点对整体的贡献是“线性”的，不需要考虑它和其他数据的乘法关系。

Geometric Mean

- 几何平均数：适合 **比例或倍数关系的数据**，比如增长率、投资回报率、浓度变化。
- 数据之间不是“独立相加”，而是 **相互作用（乘法依赖）**：一个时期的增长会影响下一个时期的基数。

SAMPLE GEOMETRIC MEAN

$$\bar{x}_g = \sqrt[n]{(x_1)(x_2)\dots(x_n)} = [(x_1)(x_2)\dots(x_n)]^{1/n} \quad (2.3)$$

1. 算数平均数的局限

- 适合加法关系
- 涉及 **比例变化、增长率或倍数** 时，算术平均数会产生误导。

2. 几何平均数的优势

- 乘法关系下的真实平均水平：几何平均数考虑了数据的连乘效应。
- 应用场景：金融（投资回报率、CAGR）、经济学（增长率）、科学（浓度、比例）

【例题】

TABLE 2.10 Percentage Annual Returns and Growth Factors for the Mutual Fund Data		
Year	Return (%)	Growth Factor
1	-22.1	0.779
2	28.7	1.287
3	10.9	1.109
4	4.9	1.049
5	15.8	1.158
6	5.5	1.055
7	-37.0	0.630
8	26.5	1.265
9	15.1	1.151
10	2.1	1.021

Computation of Geometric Mean:

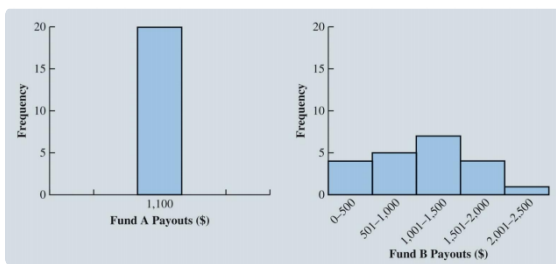
- Solution:
 - Product of the growth factors:

$$[(0.779)(1.287)(1.109)(1.049)(1.158)(1.055)(0.630)(1.265)(1.151)(1.021)] = (1.335)$$
 - Geometric mean of the growth factors: $\bar{x}_g = \sqrt[10]{1.335} = 1.029$.
 - Conclude that annual returns grew at an average annual rate of

$$(1.029 - 1) * 100\% \text{ or } 2.9\%.$$

b. Measures of Variability

Measures of Variability 变异性度量：用于描述数据的 离散程度 或 分布范围。



i. Range

- **Drawback:** Range is based on only two of the observations and thus is **highly influenced** by **extreme value**
 - 最大值 – 最小值
 - Range is based on only two of the observations and thus is **highly influenced** by extreme values. 受极端值影响

ii. Variance 方差

- **Variance** is a measure of variability that **utilizes all the data**.
- 衡量数据 离散程度的指标
- 对极端值非常敏感
- 方差的单位是原始单位的平方
 - 通常用 **标准差 (Standard Deviation)** 来解释，因为它和原始单位一致。

SAMPLE STANDARD DEVIATION

$$s = \sqrt{s^2} \quad (2.5)$$

Sample Variance

- **样本方差 (Sample Variance)**: 通常我们只有样本数据，需要用样本来估计总体方差
- **要 $n - 1$!!!**
 - 样本均值从数据里算出来的，它“消耗”了一个自由度 Degrees of Freedom

SAMPLE VARIANCE

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} \quad (2.4)$$

【例题】

Measures of Variability

Table 2.12: Computation of Deviations and Squared Deviations About the Mean for the Class Size Data

Number of Students in Class (x_i)	Mean Class Size ($\bar{x}$)	Deviation About the Mean ($x_i - \bar{x}$)	Squared Deviation About the Mean ($(x_i - \bar{x})^2$)
46	44	2	4
54	44	10	100
42	44	22	4
46	44	2	4
32	44	-12	144
		0	256
		$\Sigma(x_i - \bar{x})$	$\Sigma(x_i - \bar{x})^2$

- Computation of Sample Variance: $s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} = \frac{256}{4} = 64.$

Population Variance

- **总体方差 (Population Variance)**: 如果有整个总体的数据，分母用 N 。

- Population variance: $\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}.$

iii. Coefficient of Variation (CV)

算术平均数告诉你“平均水平”。

标准差告诉你“波动大小”。

CV告诉你“波动大小占平均水平的比例”。

The **coefficient of variation** is a descriptive statistic that indicates how large the standard deviation is relative to the mean

COEFFICIENT OF VARIATION

$$\left(\frac{\text{Standard deviation}}{\text{Mean}} \times 100 \right) \% \quad (2.6)$$

CV 越小越稳定，越大越不稳定

【例题】

Measures of Variability

Computation of Coefficient of Variation:

Illustration:

- Consider the class size data:

46 54 42 46 32

- Mean, $\bar{x} = 44$.

- Standard deviation, $s = 8$.

- Coefficient of variation = $\left(\frac{8}{44} \times 100 \right) \% = 18.2\%$.

- The coefficient of variation tells that the sample standard deviation is **18.2%** of the value of the sample mean.

COEFFICIENT OF VARIATION (CV)	
Population formula	Sample formula
$c_v = \frac{\sigma}{\mu}$	$\hat{c}_v = \frac{s}{\bar{x}}$

c. Analyzing Distributions

i. Percentiles 百分位数

- Percentile 是衡量数据分布位置的指标
- 百分位数是一个值，它表示在数据集中有多少比例的观测值 小于或等于该值。
- 第 p 百分位数：
 - 约有 $p\%$ 的数据值小于它
 - 约有 $(100 - p)\%$ 的数据值大于它

Location of the p th Percentile

$$L_p = \frac{p}{100}(n + 1) \quad (2.7)$$

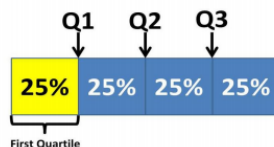
ii. Quartiles 四分位数

- **Quartiles:** When the data is divided into four equal parts:
 - Each part contains **approximately 25%** of the observations.
 - Division points are referred to as quartiles.

Q_1 = first quartile, or 25th percentile.

Q_2 = second quartile, or 50th percentile (also the median).

Q_3 = third quartile or 75th percentile.



- **Interquartile Range (IQR)**

- $IQR = Q_3 - Q_1$
- 表示中间 50% 数据的分布范围
- 常用于箱线图 (Boxplot) 中识别异常值

iii. z-Scores

- The **z-score** measures the **relative location** of a value in the data set.
 - 它回答的问题是：这个值在整个数据分布中处于什么位置？
- Helps to determine **how far** a particular value is from the mean relative to the data set's standard deviation.
- **z-score** Often called the standardized value, 表示某个数据点距离平均值有多远，并且是以 **标准差** 为单位来衡量的

公式：

$$Z = \frac{x - \mu}{\sigma}$$

其中：

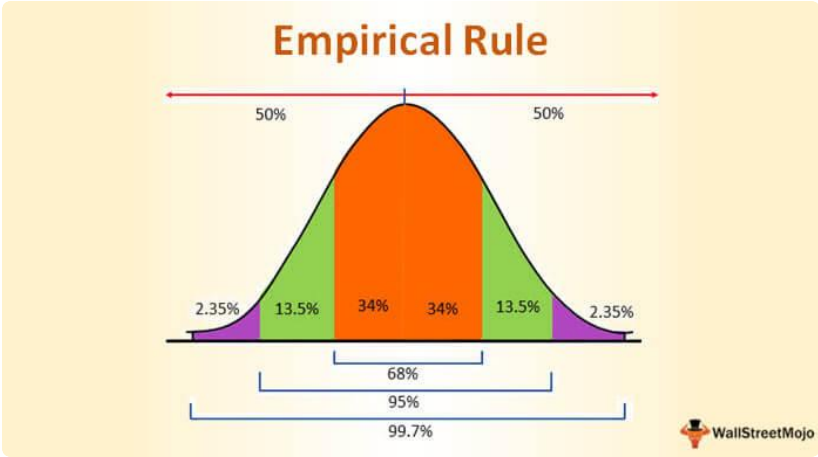
- x : 某个具体的观测值 (observation)
- μ : 总体平均值或样本均值 (mean)
- σ : 标准差 (standard deviation)

- **z-score = 数据点与平均值的距离 ÷ 标准差**
- 它是一个 **无单位的标准化指标**，用于衡量数据点在分布中的相对位置
- 越远离 0，表示越偏离平均水平
- **一般认为 $|Z| > 3$ 的数据点可能是异常值。**

iv. Empirical Rule 经验法则

When the distribution of data exhibits a symmetric bell-shaped distribution, the **empirical rule** can be used to determine the **percentage of data values** that are within a specified number of standard deviations of the mean.

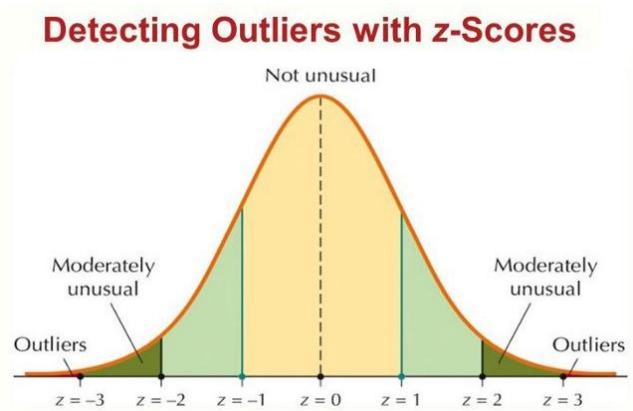
当数据呈现 正态分布时，经验法则可以用来估计数据在平均值附近的分布比例。



$\mu \pm 1\sigma$	约 68%	大多数数据在平均值上下 1 个标准差之间
$\mu \pm 2\sigma$	约 95%	几乎所有数据在平均值上下 2 个标准差之间
$\mu \pm 3\sigma$	约 99.7%	几乎全部数据都在平均值上下 3 个标准差之间

v. Identifying Outliers

- **Outliers:** Extreme values in a data set.



A **z-score less than -3 or greater than +3 is an outlier.**

Final Question Prediction

- 参考 Mid-term

• 计算题 12 marks

15. Suppose that you initially invested \$10,000 in the Stivers mutual fund, the value of each investment at the end of each subsequent year is provided in the table. What is the mean annual growth rate in the asset's value over these eight years? [12 marks]

Year	Stivers (\$)
1	11,000
2	12,000
3	13,000
4	14,000
5	15,000
6	16,000
7	17,000
8	18,000

For the Stivers mutual fund,

$$PV \times (1 + R) \times (1 + R) \times \dots \times (1 + R) = FV \quad (2 \text{ marks})$$

$$PV \times (1 + R)^8 = FV \quad (2 \text{ marks})$$

$$(1 + R)^8 = FV/PV \quad (2 \text{ marks})$$

$$1 + R = \left(\frac{FV}{PV} \right)^{\frac{1}{8}} \quad (2 \text{ marks})$$

$$18000/10000=1.8$$

$$\bar{x}_g = \sqrt[8]{(x_1)(x_2) \dots (x_8)} = \sqrt[8]{1.80} = 1.07624 \quad (2 \text{ marks})$$

15. Suppose that you initially invested \$5,000 in the Trippi mutual fund, the value of each investment at the end of each subsequent year is provided in the table. [16 marks]

Year	Trippi (\$)
1	5,600
2	6,300
3	6,900
4	7,600
5	8,500
6	9,200
7	9,900
8	10,600

a. What is the mean annual growth rate in the asset's value over these eight years? [12 marks]

b. Please use decimal scaling to normalize the mean value in part a. [4 marks]

For the Trippi mutual fund we have:

$$PV \times (1 + R) \times (1 + R) \times \dots \times (1 + R) = FV \quad (2 \text{ marks})$$

$$PV \times (1 + R)^8 = FV \quad (2 \text{ marks})$$

$$(1 + R)^8 = FV/PV \quad (2 \text{ marks})$$

$$1 + R = \left(\frac{FV}{PV} \right)^{\frac{1}{8}} \quad (2 \text{ marks})$$

$$10600/5000=2.12$$

$$10,600 = 5,000 \left[(x_1)(x_2) \dots (x_8) \right], \text{ so } [(x_1)(x_2) \dots (x_8)] = 2.12 \text{ and} \quad (2 \text{ marks})$$

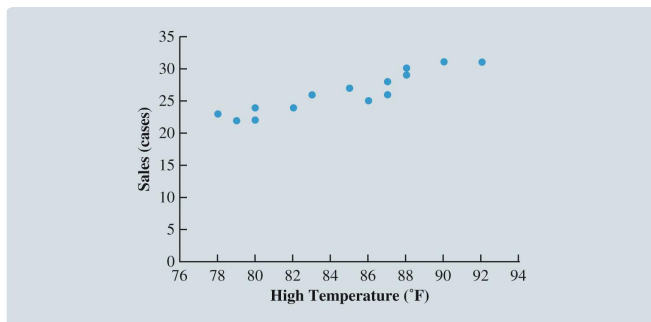
$$\bar{x}_g = \sqrt[8]{(x_1)(x_2) \dots (x_8)} = \sqrt[8]{2.12} = 1.09848$$

So the mean annual return for the Trippi mutual fund is $(1.09848 - 1)100\% = 9.848\%$. (2 marks)

6. Measures of associations between two variables

a. Scatter Charts 散点图

- 散点图是一种用 点 来 analyzing the relationship between two variables
- 每个点代表一个观测值的 两个变量坐标：横轴一个变量，纵轴另一个变量。
- 它能帮助我们识别变量之间是否存在 相关性、趋势或异常值



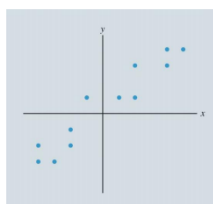
b. Covariance 协方差

- Covariance** is a descriptive measure of the **linear association** between two variables 衡量两个变量之间 线性关系方向
- 它回答的问题是：两个变量是否一起变化？如果是，它们是同向还是反向？

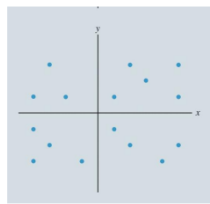
$\text{Cov}(X, Y) > 0$	两变量 正相关 Positively related
$\text{Cov}(X, Y) < 0$	两变量 负相关 Negatively related

$$\text{Cov}(X,Y) \approx 0$$

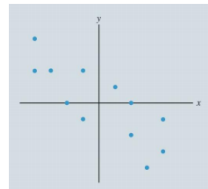
两变量之间没有明显线性关系 Not linearly related



(a)
 s_{xy} **Positive:**
(x and y are positively linearly related)



(b)
 s_{xy} **Approximately 0:**
(x and y are not linearly related)



(c)
 s_{xy} **Negative:**
(x and y are negatively linearly related)

i. Sample covariance

- 抽样调查 – 用样本估计总体

Sample Covariance

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1} \quad (2.9)$$

【例题】

Table 2.15: Sample Covariance Calculations for Daily High Temperature and Bottled Water Sales at Queensland Amusement Park

The covariance is **positive**, indicating that higher temperatures (x) are associated with higher sales (y).

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
78	23	-6.6	-3.3	21.78
79	22	-5.6	-4.3	24.08
80	24	-4.6	-2.3	10.58
80	22	-4.6	-4.3	19.78
82	24	-2.6	-2.3	5.98
83	26	-1.6	-0.3	0.48
85	27	0.4	0.7	0.28
86	25	1.4	-1.3	-1.82
87	28	2.4	1.7	4.08
87	26	2.4	-0.3	-0.72
88	29	3.4	2.7	9.18
88	30	3.4	3.7	12.58
90	31	5.4	4.7	25.38
92	31	7.4	4.7	34.78
Totals	1,185	368	-0.2	166.42

$\bar{x} = 84.6$
 $\bar{y} = 26.3$

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1} = \frac{166.42}{14 - 1} = 12.8$$

ii. Population covariance

$$\sigma_{xy} = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

iii. The Problem: Unit Dependency

- 协方差反映方向和强度，但受单位影响，不适合横向比较

One problem with using covariance is that the **magnitude** of the covariance value is **difficult to interpret**.

- Larger S_{xy} values do **not necessarily mean a stronger linear relationship** because the units of covariance depend on **the units of x and y** .

The Problem: Unit Dependency

The magnitude is dependent on units, making it hard to interpret the strength of the relationship.

The Solution



Use the **Correlation Coefficient (r)**, which is unitless.

- 协方差的值受两个变量的单位影响：
 - 如果 X 是“小时”， Y 是“分数”，协方差单位是“小时 × 分数”
 - 如果把小时换成分钟，协方差值会变成原来的 60 倍
- 所以：协方差的数值不能直接比较不同变量或不同数据集的关系强弱

c. Correlation Coefficient

- 判断变量关系强弱
- Covariance → Correlation Coefficient
- The **correlation coefficient** measures the relationship between two variables.
- **Not affected** by the units of measurement for x and y .

Sample Correlation Coefficient

$$r_{xy} = \frac{s_{xy}}{s_x s_y} \quad (2.10)$$

where

- r_{xy} = sample correlation coefficient
- s_{xy} = sample covariance
- s_x = sample standard deviation of x
- s_y = sample standard deviation of y

- 相关系数是协方差的标准化版本，无单位、范围固定、可比较性强：无单位、范围在 $[-1, +1]$

Interpretation of Correlation Coefficient:

$$-1 \leq r \leq +1$$

<i>r</i> value	Relationship between the <i>x</i> and <i>y</i> variables
< 0	Negative linear
Near 0	No linear relationship
> 0	Positive linear

【例题】

Computation of Correlation Coefficient:

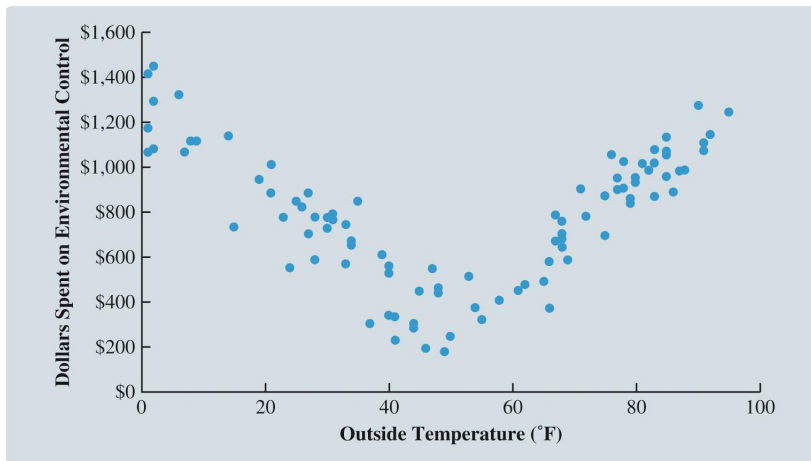
For example:

- To determine the sample correlation coefficient for bottled water sales at Queensland Amusement Park:

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{12.8}{(4.36)(3.15)} = 0.93$$

- There is a very **strong positive linear relationship** between high temperature and sales.

Problem: Non-linear relationship



- Nonlinear Relationship Producing a Correlation Coefficient Near Zero

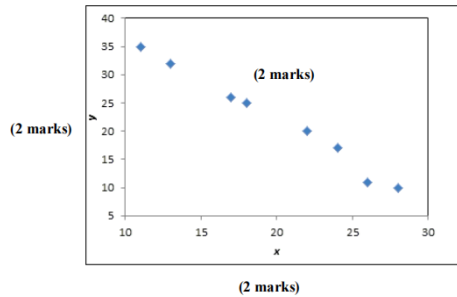
Final Question Prediction

- 参考 Mid-term
- 计算题 (20 marks)

17. Eight observations taken for two variables are as follows: [19 marks]

x_i	y_i
11	35
13	32
17	26
18	25
22	20
24	17
26	11
28	10

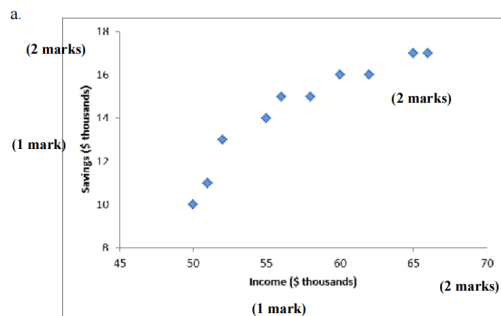
- Develop a scatter diagram with x on the horizontal axis. [6 marks]
- What does the scatter diagram developed in part (a) indicate about the relationship between the two variables? [2 marks]
- Compute and interpret the sample covariance. [7 marks]
- Compute and interpret the sample correlation coefficient. [4 marks]



17. Consider the following data on income and savings of a sample of residents in a locality: [20 marks]

Income (\$ thousands)	Savings (\$ thousands)
50	10
51	11
52	13
55	14
56	15
58	15
60	16
62	16
65	17
66	17

- Develop a scatter diagram with x on the horizontal axis. [8 marks]
- What does the scatter diagram developed in part (a) indicate the relationship between the two variables? [2 marks]
- Compute and interpret the sample covariance. [7 marks]
- Compute and interpret the sample correlation coefficient. [3 marks]



b. There appears to be a negative linear relationship between the x and y variables (2 marks)

c.

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
11	35	-8.88	13	-115.38
13	32	-6.88	10	-68.75
17	26	-2.88	4	-11.50
18	25	-1.88	3	-5.63
22	20	2.13	-2	-4.25
24	17	4.13	-5	-20.63
26	11	6.13	-11	-67.38
28	10	8.13	-12	-97.50
				-391

(1 mark) $\bar{x} = 19.88$

(1 mark) $\bar{y} = 22$

$$s_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n-1} = \frac{-391}{7} = -55.86 \quad (2 \text{ marks})$$

The negative covariance confirms that there is a negative linear relationship between the x and y variables in this data set.

d. $s_x = 6.13, s_y = 9.17$ (2 marks)

Then the correlation coefficient is calculated as:

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{-55.86}{(6.13)(9.17)} = -0.99 \quad (2 \text{ marks})$$

The correlation coefficient again confirms and indicates a strong negative linear association between the x and y variables in this data set.

b. There is a strong positive relationship between income and savings.

(2 marks)

c.

x_i	y_i	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
50	10	-7.5	-4.4	56.25	19.36	33
51	11	-6.5	-3.4	42.25	11.56	22.1
52	13	-5.5	-1.4	30.25	1.96	7.7
55	14	-2.5	-0.4	6.25	0.16	1
56	15	-1.5	0.6	2.25	0.36	-0.9
58	15	0.5	0.6	0.25	0.36	0.3
60	16	2.5	1.6	6.25	2.56	4
62	16	4.5	1.6	20.25	2.56	7.2
65	17	7.5	2.6	56.25	6.76	19.5
66	17	8.5	2.6	72.25	6.76	22.1
				292.5	52.4	116

(1 mark) (1 mark) (1 mark) (1 mark) (1 mark)

$$s_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n-1} = \frac{116}{9} = 12.89. \quad (2 \text{ marks})$$

$$d. s_x = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{292.5}{9}} = 5.70. \quad (1 \text{ mark})$$

$$s_y = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}} = \sqrt{\frac{52.4}{9}} = 2.41. \quad (1 \text{ mark})$$

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{12.89}{(5.70)(2.41)} = 0.938 \quad (1 \text{ mark})$$

7. Time Series Analysis and Forecasting

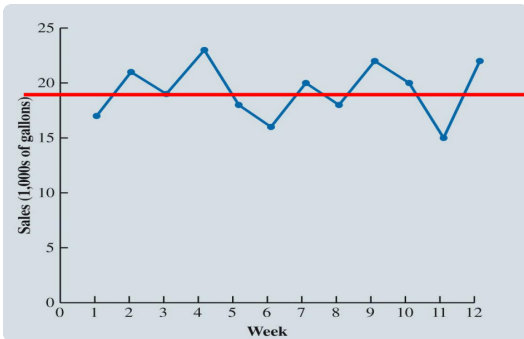
a. Time Series Patterns

- Time series:** A sequence of observations on a variable measured at successive 连续 points in time or over successive periods of time
- To identify the underlying pattern in the data, a useful first step is to construct a time series plot

- Horizontal axis: **Time**
- Vertical axis: Values of the **time series variable**

i. Horizontal Pattern

- Exists when the data **fluctuate randomly around a constant mean over time**.

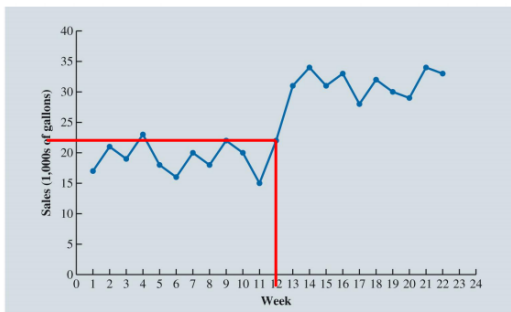


Stationary time series 平稳时间序列 is one whose statistical properties are independent of time 统计特性不随时间变化

- The process generating the data has a constant mean. 数据生成过程的均值是恒定的
- The variability of the time series is constant over time. 序列的波动性（方差）在整个时间内保持不变。

数据的平均水平和波动幅度一直稳定，不会因为时间推移而改变。

业务条件的变化可能会让时间序列整体水平发生转移。



- **平稳序列**：均值和方差不随时间变化。
- **水平转移 (Shift in Level)**：外部事件导致均值突然改变。
- **案例意义**：时间序列分析不仅要看趋势和波动，还要考虑外部事件对数据的影响。

ii. Trend Pattern

- 指时间序列中的长期变化，表现为**持续的上升或下降**。
- 它反映的是结构性的长期因素，而不是短期的随机波动。

A trend is usually the result of long-term factors such as:

- Population increases or decreases.
- Shifting demographic characteristics of the population.
- Improving technology.
- Changes in the competitive landscape.
- Changes in consumer preferences.

▼ 解释

Population increases or decreases

- 人口增长会推动消费需求，人口减少则可能导致市场萎缩。

Shifting demographic characteristics of the population 人口结构特征的变化。

- 老龄化社会会增加医疗需求，年轻人口增加则可能推动教育和娱乐产业。

Improving technology 技术进步。

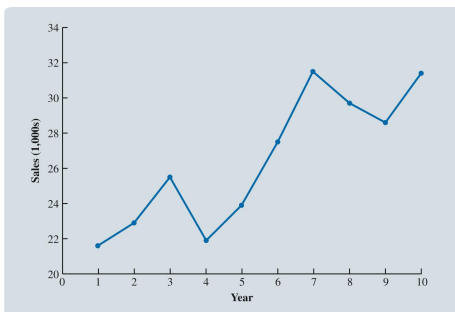
- 新技术提升生产效率，降低成本，推动长期增长。

Changes in the competitive landscape 竞争格局的变化。

- 新进入者或行业整合会改变市场份额，形成长期趋势。

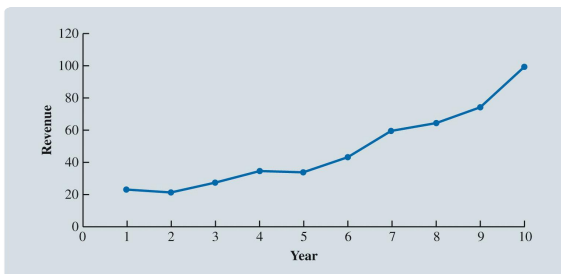
Changes in consumer preferences 消费者偏好的变化。

- 环保意识增强会推动绿色产品的长期需求。



总体上升趋势：

Visual inspection of the time series plot shows some **up-and-down** movement over the past 10 years, but the time series seems also to have a systematically **increasing, or upward, trend**.



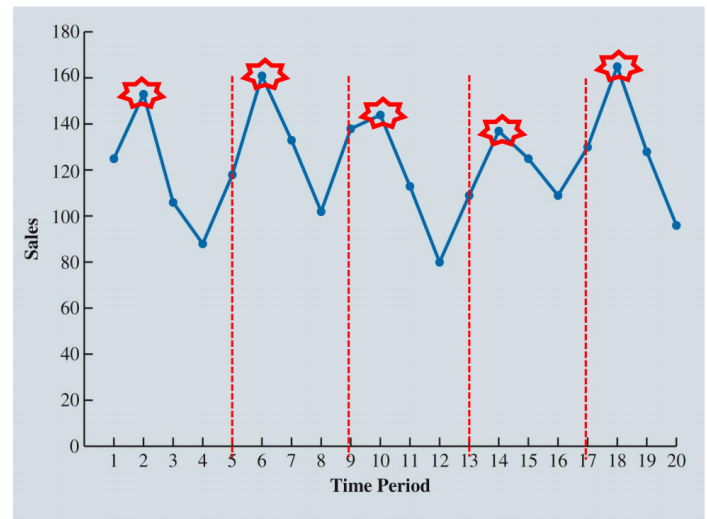
Non-linear 增长：

The time series increases in a **nonlinear** fashion; that is, the rate of change of revenue does not increase by a constant amount from one year to the next. 增长速度不是固定的

iii. Seasonal Pattern

- **Seasonal patterns are recurring patterns** over successive periods of time. 指在连续的时间周期中反复出现的规律性模式
 - A retailer selling bathing suits expects low sales in fall and winter, and peak sales in spring and summer every year.
- Seasonal patterns can also occur within shorter cycles, e.g., daily traffic volume shows peaks at rush hour, moderate flow during the day, and low flow at night. 季节性不仅限于一年周期，也可能出现在更短周期，比如一天内的交通流量：早晚高峰最高，中午适中，午夜到凌晨最低

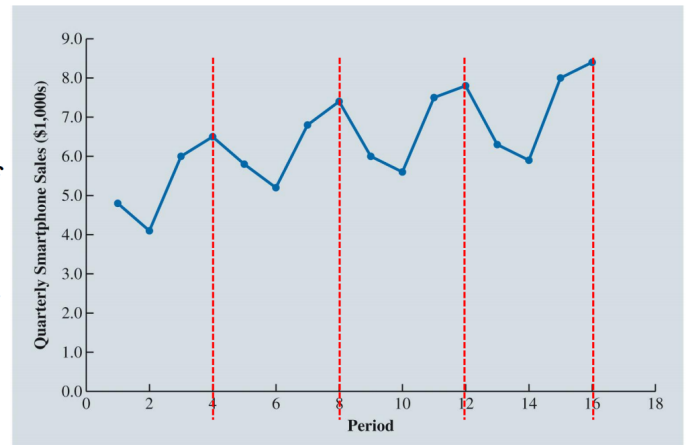
- The time series plot does not indicate a long-term trend in sales: a horizontal pattern with random fluctuation. ❌
- The first and third quarters have moderate sales, the second quarter has the highest sales, and the fourth quarter has the lowest sales volume.



iv. Trend and Seasonal Pattern

- Some time series include both a trend and a seasonal pattern

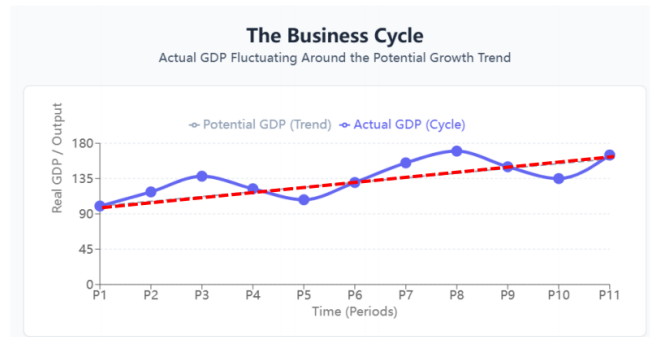
- Clearly an increasing trend is present.
- Sales are lowest in the second quarter of each year and highest in quarters 3 and 4.
- A seasonal pattern also exists for smartphone sales.



v. Cyclical Pattern

A cyclical pattern 周期性模式 exists if the time series plot shows an alternating sequence of points below and above the trendline that lasts for more than one year. 如果时间序列图在趋势线的上下方交替波动，并且这种波动持续超过一年，就说明存在周期性模式。

- In macroeconomics, while an economy generally **grows over the long term (the secular trend)**, it experiences **short-term oscillations** consisting of booms and busts.
- These **fluctuations occur around the "Potential GDP" or the long-term growth trendline.**



- Cyclical effects are often combined with long-term trend effects and referred to as **trend-cycle effects**.

- 周期性效应通常与长期趋势效应结合在一起。
- 两者合并后称为 **趋势-周期效应**。

b. Forecast Accuracy

i. Method

- **Naïve forecasting method:** Using the most recent data to predict future data
- **Average of Past Values method:** use the average of all the historical data available as the forecast for the next period.
- **Moving Averages:** Uses the average of the most recent k data values in the time series as the forecast for the next period.
- **Exponential Smoothing:** Uses a **weighted average** of past time series values as a forecast.

【详解】

1. Moving Averages

MOVING AVERAGE FORECAST

$$\begin{aligned}\hat{y}_{t+1} &= \frac{\Sigma(\text{most recent } k \text{ data values})}{k} = \frac{\sum_{i=t-k+1}^t y_i}{k} \\ &= \frac{y_{t-k+1} + \dots + y_{t-1} + y_t}{k}\end{aligned}\quad (8.6)$$

where

$\hat{y}_{t+1}$ = forecast of the time series for period $t + 1$
 y_t = actual value of the time series in period t
 k = number of periods of time series data used to generate the forecast

a small value of k is preferred

- A smaller value of k will **track shifts** in a time series more quickly
- Larger values of k will be **more effective in smoothing out random fluctuations**

- 用最近 k 个观测值的平均数来预测下一期。
- 能平滑掉短期波动。The time series curve after moving average becomes **smoother** and the fluctuation amplitude is smaller

- 适合没有明显趋势或季节性的序列。
- We will use a **three-week moving average** ($k = 3$).

Table 8.9: Summary of **Three-Week Moving Average** Calculations

Week	Time Series Value	Forecast	Forecast Error	Absolute Value of Forecast Error	Squared Forecast Error	Percentage Error	Absolute Value of Percentage Error
1	17						
2	21						
3	19						
4	23	19	4	4	16	17.39	17.39
5	18	21	-3	3	9	-16.67	16.67
6	16	20	-4	4	16	-25.00	25.00

e.g. $\hat{y}_4 = \text{average for weeks 1 to 3} = (17 + 21 + 19)/3 = 19$

Table 8.9: Summary of **Three-Week Moving Average** Calculations

Week	Time Series Value	Forecast	Forecast Error	Absolute Value of Forecast Error	Squared Forecast Error	Percentage Error	Absolute Value of Percentage Error
7	20	19	1	1	1	5.00	5.00
8	18	18	0	0	0	0.00	0.00
9	22	18	4	4	16	18.18	18.18
10	20	20	0	0	0	0.00	0.00
11	15	20	-5	5	25	-33.33	33.33
12	22	19	3	3	9	13.64	13.64
		Totals	0	24	92	-20.79	129.21

e.g. $\hat{y}_{13} = \text{average for weeks 10 to 12} = \frac{20 + 15 + 22}{3} = 19$

2. Exponential Smoothing

- **Exponential smoothing** uses a **weighted average** of past time series values as a forecast.

EXPONENTIAL SMOOTHING FORECAST

$$\hat{y}_{t+1} = \alpha y_t + (1 - \alpha) \hat{y}_t \quad (8.7)$$

$\hat{y}_{t+1}$ = forecast of the time series for period $t + 1$
 y_t = actual value of the time series in period t
 $\hat{y}_t$ = forecast of the time series for period t
 α = smoothing constant ($0 \leq \alpha \leq 1$)

- **Smoothing constant** (α) is the weight given to the actual value in period t ; weight given to the forecast in period t is $1 - \alpha$.

- Smoothing constant 平滑常数 ($0 \leq \alpha \leq 1$)
 - α 越大, 预测越“敏感”, 更快跟随最新数据变化。
 - α 越小, 预测越“平滑”, 更稳定但反应较慢。

- 指数平滑就是“把最新的实际值和之前的预测值混合”，最近的数据权重更高，过去的的数据权重逐渐减小。

Week	Time Series Value	Forecast	Forecast Error	Squared Forecast Error
1	17			
2	21	17.00	4.00	16.00
3	19	17.80	1.20	1.44
4	23	18.04	4.96	24.60
5	18	19.03	-1.03	1.06
6	16	18.83	-2.83	8.01

e.g.

• with $y_1 = 17$, we set $\hat{y}_2 = 17$ to initiate the computations.

• an actual time series value in period 2 of $y_2 = 21$.

$$\hat{y}_3 = 0.2y_2 + 0.8\hat{y}_2 = 0.2(21) + 0.8(17) = 17.8$$

Table 8.10: Summary of the Exponential Smoothing Forecasts and Forecast Errors for the Gasoline Sales Time Series with Smoothing Constant $\alpha = 0.2$

Week	Time Series Value	Forecast	Forecast Error	Squared Forecast Error
7	20	18.26	1.74	3.03
8	18	18.61	-0.61	0.37
9	22	18.49	3.51	12.32
10	20	19.19	0.81	0.66
11	15	19.35	-4.35	18.92
12	22	18.48	3.52	12.39
		Totals	10.92	98.80

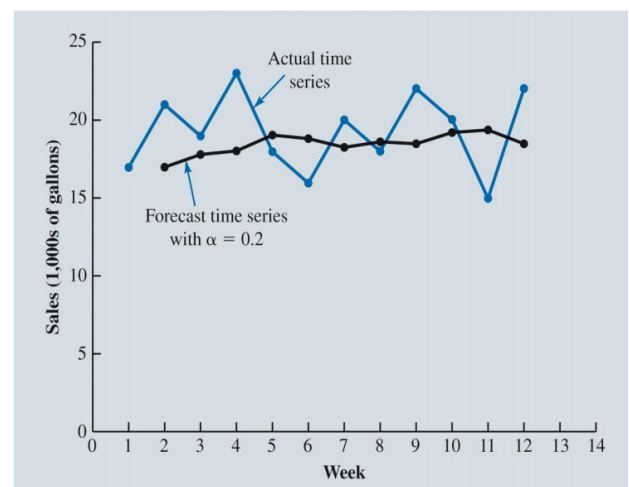
e.g.

• For week 12, we have $y_{12} = 22$ and $\hat{y}_{12} = 18.48$.

• a forecast for week 13:

$$\hat{y}_{13} = 0.2y_{12} + 0.8\hat{y}_{12} = 0.2(22) + 0.8(18.48) = 19.18$$

- Because exponential smoothing **smooths out** irregular fluctuations or random volatility, line charts become smoother.



• Meaning of Choosing α

- 如果时间序列数据 随机波动很大，就要选 较小的 α ，这样预测更平滑，不会被噪声影响太多。

- 如果数据 **变化趋势明显**，就要选 **较大的** α ，预测能更快跟上最新变化。
- 最理想的办法是选择能让 **均方误差 (MSE)** 最小的 α 。

ii. Measures

1. Forecast error

Difference between the **actual** and the forecasted values for period t .

FORECAST ERROR

$$e_t = y_t - \hat{y}_t \quad (8.1)$$

y_t and $\hat{y}_t$ as the **actual** and **forecasted** values of the time series for period t ,

Week	Time Series Value	Forecast	Forecast Error
1	17		
2	21	17	4
3	19	21	-2
4	23	19	4
5	18	23	-5
6	16	18	-2

e.g. $e_2 = y_2 - \hat{y}_2 = 21 - 17 = 4$

A **positive error**: the forecasting **underestimated** the actual value

A **negative error**: the forecast **overestimated** the actual value

2. Mean forecast error (MFE).

Mean or average of the forecast errors

MEAN FORECAST ERROR (MFE)

$$MFE = \frac{\sum_{t=k+1}^n e_t}{n - k}$$

- n = Number of periods in time series
- k = Number of periods at the beginning of the time series for which we cannot produce a naïve forecast

Week	Time Series Value	Forecast	Forecast Error
1	17		
2	21	17	4
3	19	21	-2
4	23	19	4
11	15	20	-5
12	22	15	7
	Totals		5

e.g.

- The sum of the forecast errors for the gasoline sales time series is 5.
- The mean or average error is $5/11 = 0.45$.
- The mean forecast error is positive, implying that the method is generally **under-forecasting**

3. Mean absolute error (MAE)

Mean Absolute Error (MAE): referred to as the mean absolute deviation (MAD).

它避免了正负预测误差相互抵消的问题

MEAN ABSOLUTE ERROR (MAE)

$$MAE = \frac{\sum_{t=k+1}^n |e_t|}{n - k}$$

$MAE \geq 0$, with smaller values indicating more accurate predictions.

$MAE = 0$: perfect prediction (very rare).

Week	Time Series Value	Forecast	Forecast Error	Absolute Value of Forecast Error
1	17			
2	21	17	4	4
3	19	21	-2	2
11	15	20	-5	5
12	22	15	7	7
	Totals		5	41

e.g.

- The sum of the absolute values of the forecast errors is 41.
- $MAE = \text{average of the absolute value of the forecast errors} = 41/11 = 3.73$

4. Mean squared error (MSE)

Mean Squared Error (MSE): Measure that avoids the problem of positive and negative errors offsetting each other is obtained by computing the average of the squared forecast errors.

MEAN SQUARED ERROR (MSE)

$$MSE = \frac{\sum_{t=k+1}^n e_t^2}{n - k}$$

$MSE \geq 0$, with smaller values indicating more accurate predictions.

$MSE = 0$: perfect prediction (very rare).

Week	Time Series Value	Forecast	Forecast Error	Absolute Value of Forecast Error	Squared Forecast Error
1	17				
2	21	17	4	4	16
3	19	21	-2	2	4
11	15	20	-5	5	25
12	22	15	7	7	49
	Totals		5	41	179

e.g.

- The sum of the squared errors is 179.
- $MSE = \text{average of the square of the forecast errors} = 179/11 = 16.27$.

5. Mean absolute percentage error (MAPE)

为什么需要 MAPE?

- The size of MAE or MSE depends on the **scale** of the data.
- It is difficult to compare forecasts **across different time intervals**
 - Monthly vs. Weekly sales.
- It is also difficult to compare **across different time series**
 - Gasoline vs. Oil filters. 汽油销量 vs. 滤油器销量

Mean Absolute Percentage Error (MAPE): Average of the absolute value of percentage forecast errors.

MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)

$$\text{MAPE} = \frac{\sum_{t=k+1}^n \left| \left(\frac{e_t}{y_t} \right) 100 \right|}{n - k}$$



The sum of the absolute values of the percentage errors

- MAPE 就是“平均预测误差占实际值的百分比”
- The closer to 0%, the better. MAPE = 0% indicates a perfect forecast (very rare)

Week	Time Series Value	Forecast	Forecast Error	Absolute Value of Forecast Error	Squared Forecast Error	Percentage Error	Absolute Value of Percentage Error
1	17						
2	21	17	4	4	16	19.05	19.05
3	19	21	-2	2	4	-10.53	10.53
4	23	19	4	4	16	17.39	17.39
5	18	23	-5	5	25	-27.78	27.78
6	16	18	-2	2	4	-12.50	12.50
7	20	16	4	4	16	20.00	20.00
8	18	20	-2	2	4	-11.11	11.11
9	22	18	4	4	16	18.18	18.18
10	20	22	-2	2	4	-10.00	10.00
11	15	20	-5	5	25	-33.33	33.33
12	22	15	7	7	49	31.82	31.82
Totals			5	41	179	1.19	211.69

e.g.

- the sum of the absolute values of the percentage errors is $\sum_{t=k+1}^{12} \left| \left(\frac{e_t}{y_t} \right) 100 \right| = 211.69$
- the MAPE, which is the average of the absolute value of percentage forecast errors, is $211.69/11 = 19.24\%$

Final Question Prediction

- Short Q 和计算都有可能